

A Multi-Vendor Corpus of Reference Low-Dose CT Reconstructions with Calibrated Per-Pixel Uncertainty and Signal-Equivalence Credentials

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Abstract

Low-dose CT reconstruction research is rich in methods but poor in *shared, reusable derived data*: published methods release a model or a number, rarely the per-case reconstructed volumes, the per-pixel uncertainty that a clinical user would triage on, or a machine-checkable certificate of diagnostic equivalence. We describe a deposited corpus that supplies exactly this layer on top of the multi-vendor PWM-LDCT benchmark ([to be confirmed, 2027a](#)). For every test scan, at four dose levels (10%, 25%, 50%, 100%) and across multiple CT vendors, the corpus provides: (i) a **reference reconstruction** produced by an open, fully specified unrolled-iterative pipeline; (ii) a **per-pixel uncertainty map** from a five-model deep ensemble; (iii) the **paired per-pixel error map** against the full-dose reference, enabling calibration studies without re-running any model; (iv) **downstream-task output maps** (lung-nodule detection scores); (v) machine-readable **signal-equivalence credentials** computed under the framework of [to be confirmed \(2027b\)](#); and (vi) the matched reconstructions of five baseline methods for comparison. Every record is content-addressed, released under CC BY 4.0, and accompanied by an Apache-2.0 generating pipeline packaged as a pinned Docker RunBundle so the corpus is regenerable bit-for-bit within floating-point tolerance. Technical validation reports reconstruction fidelity, cross-vendor consistency, and—centrally for reuse—the agreement between the released uncertainty maps and the released error maps (Spearman ρ). The corpus is intended for reuse in uncertainty-calibration research, cross-vendor robustness studies, downstream-task evaluation under dose reduction, and as a stable comparison baseline for new reconstruction methods.

1 Background & Summary

Deep-learning low-dose CT reconstruction has produced a decade of method papers reporting fractional-dB gains ([Chen et al., 2017](#); [Wang et al., 2020](#); [to be confirmed at submission, 2024](#)), yet the field shares almost no *derived* data. Two structural gaps follow. First, reconstructed volumes are seldom released: a researcher who wants to study how reconstruction error distributes across anatomy, dose, or vendor must retrain each method from scratch. Second, the two quantities most relevant to clinical deployment—*per-pixel uncertainty* and a *certificate of task-level equivalence*—are almost never published as data a third party can inspect. A deployer choosing a reconstruction for a new scanner, or a calibration researcher studying whether predicted variance tracks true error, has nothing reusable to build on.

The raw substrate for closing these gaps now exists: the PWM-LDCT benchmark ([to be confirmed, 2027a](#)) provides multi-vendor paired (full-dose, low-dose) CT, extending the field’s two single-vendor real-paired resources, AAPM-2016 ([AAPM, 2016](#)) and Mayo LDCT-PD ([McCollough et al., 2017](#)), both Siemens-only. What has been missing is the *derived layer*: a uniformly produced set of reconstructions, uncertainty maps, error maps, task outputs, and equivalence credentials over that substrate.

This Data Descriptor releases that layer. We generate every record with a single, fully specified open pipeline—an unrolled iterative reconstruction with a weight-shared denoiser and a five-model deep ensemble for uncertainty (Methods)—and deposit the outputs as a content-addressed corpus. We emphasise that the pipeline is a *means of generating reproducible data*, not the object of this paper; its architecture is reported in full only so the data are reproducible and the uncertainty maps are interpretable.

Relationship to the PWM-LDCT benchmark (WS-1). This corpus is deliberately disjoint from the underlying benchmark dataset. WS-1 releases the *raw paired CT scans*; the present corpus releases the *reconstruction, uncertainty, error, task-output, and credential records derived from them*. The two are intended to be cited together: WS-1 as the input data, this corpus as the reference derived data.

Reuse value. The corpus enables, without any model retraining: (1) **uncertainty-calibration research**—the paired (uncertainty map, error map) records are a ready-made benchmark for calibration metrics and recalibration methods; (2) **cross-vendor robustness studies**—reconstructions and errors are vendor-labelled, supporting leave-one-vendor-out analyses; (3) **downstream-task evaluation under dose reduction**—released task-score maps let others study how dose affects detection without re-deriving reconstructions; (4) **credential-based comparison**—the machine-readable signal-equivalence credentials are a stable target that new methods can report against under a content-addressed framework hash; and (5) a **stable comparison baseline**—five matched baseline reconstructions accompany the reference so new methods compare against identical inputs.

Summary of what is released. Table 1 summarises the record types; section 3 gives the full file-level manifest. All records, for all scans, at all dose levels and vendors, are deposited together under a single repository DOI ([[TODO: figshare/Zenodo DOI on deposit](#)]).

2 Methods

This section specifies how each released record was generated. It is included to make the corpus reproducible and the uncertainty maps interpretable; it is not a methodological contribution of this Data Descriptor.

2.1 Input data and provenance

Inputs are the PWM-LDCT ([to be confirmed, 2027a](#)) scans at signal ratios $r \in \{0.10, 0.25, 0.50\}$, patient-level split into train/validation/test, vendor-labelled. Reference (target) images are the patient’s full-dose reconstruction. No scan crosses the train/test boundary at the patient level; all released records are for the held-out test split.

Low-dose provenance (real-paired vs simulated). Each (vendor, dose) stratum is one of two kinds, recorded per record in `scan_meta.json` (`low_dose_origin` $\in \{\text{real_paired}, \text{simulated}\}$): *real-paired* acquisitions are genuine repeat low-dose scans from the source benchmark ([to be confirmed, 2027a](#)); *simulated* low-dose is generated from the full-dose projections by Poisson photon-count reduction in the `ct_radon` forward model ([PWM Protocol, 2026](#)) at the stated r . Reusers must not treat the two as interchangeable: simulated strata characterise algorithmic behaviour under an idealised noise model, real-paired strata characterise behaviour under true acquisition physics. The per-stratum breakdown is reported in Table 3 and carried in every record’s metadata.

The $r = 1.00$ level. Full dose is the reference, not a credentialed dose level. Where an $r = 1.00$ reconstruction is released it is the reference image x^{ref} itself; no error map is emitted at $r = 1.00$ (the error against the reference is identically zero) and no credential is defined there (signal-equivalence credentials compare a *reduced*-dose method against the full-dose reference). All released error maps, task-score maps, and credentials are therefore at $r \in \{0.10, 0.25, 0.50\}$.

Ethics and human-subjects provenance. This corpus contains no new human-subjects data: every record is a derived image or a numerical summary computed from the PWM-LDCT benchmark (to be confirmed, 2027a), which governs the underlying acquisitions, their institutional-review-board approvals, and informed-consent basis. The released NIfTI volumes carry no protected health information: voxel arrays only, with all DICOM patient-identifying header fields removed at generation, and `scan_id` is the WS-1 pseudonymous identifier (never an institutional medical-record number). Redistribution of these derived reconstructions under CC BY 4.0 is permitted by the WS-1 data-use terms; reusers remain bound by those terms for any attempt to re-link records to source acquisitions. No date-of-service, geographic, or demographic field finer than the anatomy/vendor labels in `scan_meta.json` is released.

2.2 Reconstruction pipeline

Reconstructions are produced by an unrolled iterative scheme (Adler and Öktem, 2018; Mardani et al., 2018). Let $\mathcal{R} : \mathbb{R}^{H \times W} \rightarrow \mathbb{R}^{n_{\text{views}} \times n_{\text{dets}}}$ be the Radon forward operator and $\mathcal{R}^\dagger$ its filtered back-projection adjoint, both taken from the open `pwm_core.contrib.modalities.ct_radon` implementation (PWM Protocol, 2026) validated against the PWM modality test suite; we do not re-implement them, so any user regenerating the corpus inherits the identical, version-pinned forward model.

Given a low-dose sinogram y , the reconstruction $\hat{x}$ is computed by $K = 10$ data-consistency / denoising iterations (Algorithm 1). The denoiser f_θ is a U-Net (≈ 4 M parameters, four encoder / four decoder stages with (32, 64, 128, 256) channels, GELU, GroupNorm) whose weights are shared across iterations; the step size τ is a learned scalar initialised at $0.5/\|\mathcal{R}\|_2^2$.

Algorithm 1 Unrolled iterative reconstruction (per ensemble member)

```

1:  $x^{(0)} \leftarrow \text{TV-FBP}(y)$  ▷ TV-regularized warm start
2: for  $k = 0, 1, \dots, K-1$  do
3:    $x^{(k+1/2)} \leftarrow x^{(k)} - \tau \cdot \mathcal{R}^\dagger(\mathcal{R}(x^{(k)}) - y)$  ▷ Data consistency
4:    $x^{(k+1)} \leftarrow f_\theta(x^{(k+1/2)})$  ▷ Learned denoising
5: end for
6: return  $\hat{x} \leftarrow x^{(K)}$ 

```

2.3 Uncertainty quantification

Each released reconstruction is the mean of a five-model deep ensemble (Lakshminarayanan et al., 2017): five instances of the architecture above, identical except for the random seed governing weight initialisation and data ordering (seeds {42, 43, 44, 45, 46}). For a low-dose input y , member m produces $\hat{x}_m(y)$; the released point estimate and per-pixel uncertainty are

$$\hat{x}(y) = \frac{1}{5} \sum_{m=1}^5 \hat{x}_m(y), \quad \sigma(y)_{ij} = \sqrt{\frac{1}{5} \sum_{m=1}^5 (\hat{x}_m(y)_{ij} - \hat{x}(y)_{ij})^2}. \quad (1)$$

Deep ensembles are chosen over MC dropout because they calibrate better under distribution shift (Ovadia et al., 2019; Gal and Ghahramani, 2016) and have bounded ($5\times$) inference cost. The released $\sigma(y)$ map is the quantity intended for case-level triage; its *validity* as a triage signal is established in Technical Validation by comparison with the released error map.

2.4 Derived per-record quantities

For each test scan and dose level the pipeline emits, in addition to $\hat{x}(y)$ and $\sigma(y)$: the **per-pixel error map** $e(y)_{ij} = |\hat{x}(y)_{ij} - x_{ij}^{\text{ref}}|$ against the full-dose reference, and a **lung-nodule detection score map** from a frozen pre-trained detector applied to $\hat{x}(y)$ (detector specified in Section 2.5). (Version 1 of the corpus releases this single chest / lung-nodule task; additional tasks such as abdominal liver-lesion segmentation are deferred to a future release so that the released records and the task description remain in exact correspondence.)

In addition, at the *cohort* level—one record per (task, r , subpopulation, method) stratum rather than per scan—the pipeline emits a **signal-equivalence credential** (5-tuple $(r, T, \varepsilon, \alpha, \Pi)$ with verdict and confidence interval) computed with the `pwm_dose_equivalence` library (to be confirmed, 2027b); each credential is internally audited with `pwm-audit` before it is written, and is regenerable from the released task-score arrays. The five baseline methods (Section 3) are run through the identical emit path so their records are field-for-field comparable.

2.5 Frozen downstream-task detector

The lung-nodule detection score maps are produced by a single *frozen* detector held fixed across every method, dose level, and record, so the task layer measures the diagnostic content each reconstruction preserves rather than any adaptation of the detector itself. The detector is **nnDetection** (Baumgartner et al., 2021)—a self-configuring 3D Retina U-Net—run in its published configuration without modification; the exact release is pinned in Supplementary Table S2. It is trained *once* on the **LUNA16** benchmark (Setio et al., 2017), the 888-scan LIDC/IDRI (Armato III et al., 2011) subset annotated for pulmonary nodules with $\geq 3/4$ radiologist agreement, and is never fine-tuned on PWM-LDCT reconstructions. The LUNA16 training patients are disjoint from the PWM-LDCT cohort, so the detector is an external, leakage-free evaluator applied identically to all methods.

Operating-point provenance. The credentialing operating point is fixed *before* any PWM-LDCT reconstruction is scored: on the held-out LUNA16 validation fold we take the detection-score threshold whose free-response ROC (FROC) yields a mean of 0.1 false-positive detections per scan ($\text{FPR} = 0.1$) and freeze it for all subsequent scoring. The released task-score maps are the raw, pre-threshold detector scores, so reusers may re-derive performance at any operating point; the AUC in Table 9 and the task credentials are read at this single frozen $\text{FPR} = 0.1$ point, identical across all methods and doses. The frozen weights, inference configuration, and version pin are deposited in the RunBundle (Section 6) so the task layer is independently reproducible.

2.6 Training and generation configuration

Models are trained on the PWM-LDCT training split with paired targets at $r \in \{0.10, 0.25, 0.50\}$ mixed per minibatch with weights (0.4, 0.4, 0.2); augmentation is physical-consistency-preserving only (random flip $p=0.5$; rotation $[-5^\circ, 5^\circ]$; window-level shift $[-50, 50]$ HU). Optimiser AdamW (Loshchilov and Hutter, 2019), cosine LR with 5% warmup, batch size 4 at 512×512 , single A100; CuDNN deterministic, all RNGs seed-propagated. The complete hyperparameter table, library version pins, and compute disclosure accompany the deposit (Supplementary Tables S1–S3) so the corpus is regenerable.

3 Data Records

All records are deposited at [TODO: figshare/Zenodo DOI on deposit] under CC BY 4.0, mirrored to a content-addressed PWM RunBundle on IPFS (CID in Section 6) for integrity verification. The deposit is a single archive with the directory layout below; every leaf file is additionally listed with its SHA-256 in `MANIFEST.sha256` at the archive root.

Directory layout.

```
pwm-ldct-recon-corpus/
  MANIFEST.sha256          # SHA-256 of every leaf file
  dataset_metadata.json    # ISA-style top-level metadata, DOI, licence
  reconstructions/
    <vendor>/<scan_id>/r<dd>/      # dd in {010,025,050,100}
      recon_mean.nii.gz          # reference reconstruction (this method)
      uncertainty_sigma.nii.gz   # per-pixel ensemble std dev
      error_abs.nii.gz           # |recon - full_dose_ref|
```

Record type	Format	Per-record content
Reference reconstruction	NIFTI (.nii.gz)	Ensemble-mean reconstruction $\hat{x}(y)$, HU, $512 \times 512 \times Z$
Uncertainty map	NIFTI	Per-pixel ensemble std. dev. $\sigma(y)$, same grid
Error map	NIFTI	Per-pixel $ \hat{x} - x^{\text{ref}} $ vs full dose, same grid
Task-score map	NIFTI	Frozen lung-nodule detector score map on $\hat{x}(y)$
Credential	JSON	5-tuple signal-equivalence credential (to be confirmed, 2027b) + verdict + CI + framework hash
Baseline reconstructions	NIFTI	Same fields, for 5 baseline methods (Sec. 3.1)
Per-scan metadata	JSON	Vendor, dose ratio r , anatomy, split, source-scan ID (links to WS-1)

Table 1: Overview of released record types. Counts and total size per type are in Table 4; the full directory manifest is below.

```

    task_nodule_score.nii.gz      # frozen-detector score map
    scan_meta.json               # vendor, r, anatomy, split, WS-1 source id
baselines/
  <method>/<vendor>/<scan_id>/r<dd>/ # method in {fbp,redcnn,wgan_vgg,
    recon.nii.gz                  #               transformer,diffusion}
    error_abs.nii.gz
    task_nodule_score.nii.gz
credentials/                     # cohort-level: one credential per
  <task>/r<dd>/<method>__<vendor>.json # (task x dose x method x vendor)
  all_credentials.jsonl          # flat queryable index of every credential
checksums_and_versions.json      # library version pins for regeneration

```

Naming and indexing. `scan_id` is the WS-1 source identifier, so any record back-links to its raw input. `r<dd>` encodes the dose ratio. The flat `all_credentials.jsonl` index lets a reuser query credentials (by vendor, dose, task, verdict) without walking the tree.

File-format conventions. All image records are single-file gzip-compressed NIFTI-1 (.nii.gz), float32, with an `sform` set to a scaled identity and voxel spacing carried in the header `pixdim`. The affine is an orientation-normalised scaled identity by design (patient-position and origin fields from the source DICOM are not propagated, as part of de-identification); records are therefore in a canonical voxel frame, not the original scanner frame, and should not be re-registered to source-patient coordinates. Reconstruction, reference, and error volumes are in Hounsfield units (HU); uncertainty maps are in HU (the ensemble standard deviation of the reconstruction); task-score maps are unitless detector scores in $[0, 1]$. All maps for one record share an identical voxel grid, so $\hat{x}$, σ , and e are voxel-aligned and can be stacked without resampling. JSON records are UTF-8; credentials validate against the machine-readable schema bundled with the deposit (`credential_schema.json`) and the WS-2 library (to be confirmed, 2027b).

Data dictionary. Each `scan_meta.json` carries the fields in Table 2; each `credential.json` is the WS-2 5-tuple wire format (top-level `schema_version`, `framework_hash`, and a `credential` object holding `signal_ratio`, `task`, `epsilon`, `alpha`, `subpopulation`, `estimator`, `n_test`, `delta_mean`, `delta_ci_low`, `delta_ci_high`, `verdict`, and a `sample_size_check` block), specified field-by-field in the companion credential reading guide and the bundled JSON Schema.

3.1 Baseline reconstructions included for comparison

To make the corpus a self-contained comparison resource, the matched reconstructions of five baselines are included, each run through the identical emit path: FBP; RED-CNN (Chen et al., 2017); WGAN-VGG (Yang et al., 2018); a transformer reconstruction (Wang et al., 2023); and a diffusion / score-based

Field	Type	Meaning
<code>scan_id</code>	string	WS-1 pseudonymous source identifier (back-link)
<code>vendor</code>	string	CT manufacturer (e.g. Siemens / GE / Canon)
<code>r</code>	float	signal ratio $\in (0, 1]$ (dose level)
<code>anatomy</code>	string	body region (chest in the v1 lung-nodule corpus)
<code>split</code>	string	always test for released records
<code>source_scan_id</code>	string	WS-1 series this record was reconstructed from
<code>low_dose_origin</code>	string	real_paired or simulated (see Methods)

Table 2: `scan_meta.json` fields. No field finer than anatomy/vendor is released (see Ethics and human-subjects provenance).

Vendor	Patients	Scans	Real-paired strata	Simulated strata
[TODO: vendor 1]	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]
[TODO: vendor 2]	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]
Total	[TODO: N]	[TODO: N]	[TODO: N]	[TODO: N]

Table 3: Cohort composition by vendor: distinct patients and scans, and how many (vendor, dose) strata are real-paired vs simulated low-dose. Populated at deposit (`package_corpus.py` derives the patient / scan counts from `scan_meta.json`).

reconstruction (to be confirmed at submission, 2024). Each baseline is reproduced within ≤ 0.5 dB PSNR of its published numbers before its records are emitted; reproduction recipes accompany the deposit. The baselines are single-model methods and therefore carry no **uncertainty_sigma** record (ensemble disagreement is defined only for the deep-ensemble reference method); their record set is reconstruction, error map, task-score map, and credential.

4 Technical Validation

Validation establishes that the released records are fit for reuse. It is organised around the three properties a reuser depends on: that the reconstructions are faithful, that they are consistent across vendors, and that the released uncertainty maps actually track the released error maps.

Reconstruction fidelity. Table 5 reports PSNR, SSIM, and LPIPS (Zhang et al., 2018) of every released reconstruction against its full-dose reference, per dose level, for the reference method and the five baselines. These quantify how close each released volume is to ground truth; they are descriptive data-quality evidence, not a competition.

Cross-vendor consistency. Because records are vendor-labelled, we validate that fidelity does not collapse across vendors. Table 6 reports the PSNR difference Δ under leave-one-vendor-out generation (the held-out vendor’s scans reconstructed by models trained without that vendor). This characterises the regime in which the released cross-vendor records are trustworthy for reuse versus extrapolative.

Uncertainty–error agreement (central reuse validation). The corpus’s distinctive reuse value is the paired (uncertainty, error) records, so we validate it along three complementary axes rather than one. (i) **Rank tracking:** the Spearman correlation between $\sigma(y)$ and $e(y)$, per pixel and aggregated per case (Table 7) — does larger predicted uncertainty co-occur with larger error? (ii) **Calibration magnitude:** a reliability diagram of predicted σ against observed RMS error per σ -bin (Figure 1), summarised by an expected-calibration-error (ECE), the mean absolute gap between the two — rank correlation can be high while the variance is systematically mis-scaled, which ECE catches. (iii) **Interval coverage:** the empirical fraction of voxels whose true error falls within $\pm k \sigma$ for $k \in \{1, 2, 3\}$ (Table 8); a well-calibrated Gaussian-like

Record type	Count	Format	Total size
Reference reconstructions	[TODO: N]	NIFTI	[TODO: GB]
Uncertainty maps	[TODO: N]	NIFTI	[TODO: GB]
Error maps	[TODO: N]	NIFTI	[TODO: GB]
Task-score maps	[TODO: N]	NIFTI	[TODO: GB]
Credentials	[TODO: N]	JSON	[TODO: MB]
Baseline reconstructions	[TODO: N]	NIFTI	[TODO: GB]
Total	[TODO: N]	—	[TODO: GB]

Table 4: Record counts and sizes. Populated when the corpus is generated and deposited (WS-3 Phase 3, D9+270).

Method	$r = 0.10$		$r = 0.25$		$r = 0.50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
FBP	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RED-CNN	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
WGAN-VGG	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Transformer	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Diffusion	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Reference	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 5: Reconstruction fidelity of the released records against the full-dose reference, with 95% bootstrap CIs ($B=1000$) over test patients. Populated at deposit (D9+270).

variance yields $\approx 68/95/99.7\%$. Reporting all three lets a reuser distinguish a uncertainty map that merely *looks* error from one that is *quantitatively* calibrated for interval-based triage; where they disagree, the corpus documents which property holds so the variance is not over-trusted. This is the single most important validation for the corpus’s stated calibration-research reuse.

[TODO: FIGURE: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error (RMS per σ -bin) on the held-out test split, annotated with the expected-calibration-error (ECE).]

Figure 1: Reliability diagram: released per-pixel ensemble standard deviation vs observed per-pixel reconstruction error (RMS per σ -bin) on the held-out test split, annotated with the expected-calibration-error (ECE). (placeholder; populate when the corpus is deposited).

Downstream-task validity. Table 9 reports lung-nodule detection AUC (fixed FPR = 0.1) computed from the released task-score maps with a frozen detector held constant across all methods. This validates that the released reconstructions preserve diagnostically relevant structure, and documents the known divergence between pixel fidelity and task performance (Kim et al., 2019) so reusers interpret both layers correctly.

Credential validation. Each released credential is internally audited with the `pwm-audit` tool (to be confirmed, 2027b) (schema validity, verdict-vs-CI self-consistency, framework-hash recognition); Table 10 summarises the verdicts. The audit is bundled in the deposit so a reuser can re-verify every credential without re-running the bootstrap.

Reproducibility of generation. The full corpus regenerates from the deposited Apache-2.0 RunBundle: `docker run` on a fresh machine reproduces every record bit-identically within IEEE-754 tolerance (seeds, library pins, and CuDNN-deterministic settings in Supplementary Table S2). The RunBundle is independently re-executed by the PWM challenge scoring service before posting, providing third-party regeneration evidence.

[TODO: TABLE: Cross-vendor PSNR difference (Δ , dB) under leave-one-vendor-out generation; reported so reusers know which vendor records are interpolation vs extrapolation]

Table 6: Cross-vendor PSNR difference (Δ , dB) under leave-one-vendor-out generation; reported so reusers know which vendor records are interpolation vs extrapolation (placeholder; populate when the corpus is deposited).

Dose	Spearman ρ (per-pixel)	Spearman ρ (per-case mean)	95% CI
$r = 0.10$	[TODO:]	[TODO:]	[TODO:]
$r = 0.25$	[TODO:]	[TODO:]	[TODO:]
$r = 0.50$	[TODO:]	[TODO:]	[TODO:]

Table 7: Agreement between released uncertainty maps and released error maps. Populated at deposit (D9+270).

5 Usage Notes

Loading. All image records are NIFTI and load with any standard reader (`nibabel`, `ITK`); credentials are JSON validated against the published schema ([to be confirmed, 2027b](#)). The `all_credentials.jsonl` index supports tabular queries by vendor, dose, task, and verdict. Each record’s `scan_meta.json` links back to its WS-1 raw source ([to be confirmed, 2027a](#)). A minimal read of one record’s voxel-aligned reconstruction, uncertainty, and error maps:

```
import nibabel as nib, json
base = "reconstructions/Siemens/<scan_id>/r025/"
recon = nib.load(base + "recon_mean.nii.gz").get_fdata()      # HU
sigma = nib.load(base + "uncertainty_sigma.nii.gz").get_fdata() # HU, same grid
error = nib.load(base + "error_abs.nii.gz").get_fdata()       # |recon - full_dose|
meta = json.load(open(base + "scan_meta.json"))               # vendor, r, anatomy, ...
# the (sigma, error) pair is the ready-made calibration benchmark
```

Intended reuses. (1) Uncertainty-calibration research: treat the paired (`uncertainty_sigma`, `error_abs`) maps as a ready calibration benchmark. (2) Cross-vendor robustness: use vendor labels and the leave-one-vendor-out validation to scope claims. (3) Dose-dependent downstream-task studies: use the released task-score maps without re-deriving reconstructions. (4) Method comparison: benchmark a new reconstruction against the matched baseline records on identical inputs and report a credential against the same framework hash.

Cautions for reuse. The released uncertainty is ensemble disagreement; it can be non-informative where all members fail in correlated ways (out-of-distribution anatomy)—inspect the reliability diagram before triaging on σ . Cross-vendor records for vendors not in training are extrapolation, not measurement. Dose ratios below 0.10 are not characterised. Reconstructions are single-slice 2D. Credential **INDETERMINATE** verdicts on this cohort reflect sample size ([to be confirmed, 2027b](#)), not evidence of non-equivalence, and should not be read as a negative finding about a method.

Licensing. Data records: CC BY 4.0. Generating code and RunBundle: Apache 2.0. Reuse requires citing this Data Descriptor and the WS-1 source dataset ([to be confirmed, 2027a](#)).

6 Code Availability

The complete generating pipeline—training, inference, ensemble aggregation, error/task/credential emission, and the evaluation harness that produced every Technical Validation number—is released under

Dose	ECE	$\pm 1\sigma$ cov.	$\pm 2\sigma$ cov.	$\pm 3\sigma$ cov.
$r = 0.10$	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$r = 0.25$	[TODO:]	[TODO:]	[TODO:]	[TODO:]
$r = 0.50$	[TODO:]	[TODO:]	[TODO:]	[TODO:]

Table 8: Calibration magnitude (expected-calibration-error) and empirical $\pm k\sigma$ interval coverage of the released uncertainty maps; well-calibrated variance yields coverage $\approx 68/95/99.7\%$. Populated at deposit (D9+270).

[TODO: TABLE: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level]

Table 9: Lung-nodule detection AUC (frozen detector, fixed FPR = 0.1) computed from the released task-score maps, per method, per dose level (placeholder; populate when the corpus is deposited).

Apache 2.0 at https://github.com/integritynoble/low_dose_CT/tree/main/WS-3_reference_method, and packaged as a pinned Docker RunBundle (CID [TODO: IPFS CID at deposit]; anchored by a PWM L4 cert for integrity). Pre-trained ensemble weights, the pinned Dockerfile, evaluation scripts, random seeds, and the training-compute disclosure (Supplementary Table S3) permit bit-identical regeneration of the corpus within floating-point tolerance on equivalent hardware. The forward model is the upstream `pwm_core.contrib.modalities.ct_radon` (PWM Protocol, 2026); the credentialing library is `pwm_dose_equivalence` (to be confirmed, 2027b). The credential-emission and corpus-packaging steps (record layout, `pwm-audited` credential issuance, machine-readable metadata, and the SHA-256 integrity manifest) are unit-tested, and a runnable synthetic fixture reproduces the full record-generation $\rightarrow$ credential $\rightarrow$ metadata $\rightarrow$ manifest pipeline end-to-end on miniature data, so a reviewer can exercise the corpus structure and audit chain in one command without the full dataset.

Data Citations

The deposited corpus: [TODO: figshare/Zenodo DOI on deposit] (CC BY 4.0).

Source benchmark dataset (WS-1): to be confirmed (2027a).

Signal-equivalence framework and credential schema: to be confirmed (2027b).

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[TODO: Fill at submission.]

References

- AAPM. 2016 NIH-AAPM-Mayo clinic low dose CT grand challenge. <https://www.aapm.org/grandchallenge/lowdosect/>, 2016.
- Jonas Adler and Ozan Öktem. Learned primal-dual reconstruction. *IEEE Transactions on Medical Imaging*, 37(6):1322–1332, 2018.
- S. G. Armato III, G. McLennan, L. Bidaut, et al. The LIDC/IDRI database. *Medical Physics* 38(2):915–931, 2011.
- Michael Baumgartner, Paul F. Jäger, Fabian Isensee, and Klaus H. Maier-Hein. nnDetection: A self-configuring method for medical object detection. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, pages 530–539, 2021.

r	Task	ε	α	Subpopulation	Verdict	CI
0.25	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]
0.10	Lung nodule AUC	0.05	0.05	adult-chest test split	[TODO:]	[TODO:]

Table 10: Signal-equivalence credentials (to be confirmed, 2027b) for the reference reconstruction against full-dose, audited with `pwm-audit`. Verdicts: **PASS** / **FAIL** / **INDETERMINATE**. Note: the WS-1 test cohort sits in the **INDETERMINATE**-dominated regime for the AUC default; absence of **PASS** reflects cohort size, not non-equivalence (see to be confirmed, 2027b).

- Hu Chen, Yi Zhang, Mannudeep K. Kalra, et al. Low-dose CT with a residual encoder-decoder convolutional neural network. *IEEE Transactions on Medical Imaging*, 36(12):2524–2535, 2017.
- Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.
- H. Kim et al. Diagnostic accuracy of low-dose CT with iterative reconstruction. *European Radiology*, 29: 3146–3155, 2019.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- Morteza Mardani, Qingyun Sun, David Donoho, Vardan Papyan, Hatem Monajemi, Shreyas Vasanawala, and John Pauly. Neural proximal gradient descent for compressive imaging. *Advances in Neural Information Processing Systems*, 2018.
- Cynthia H. McCollough, Adam C. Bartley, Rickey E. Carter, et al. Low-dose CT for the detection and classification of metastatic liver lesions: Results of the 2016 low dose CT grand challenge. *Medical Physics*, 44(10):e339–e352, 2017.
- Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019.
- PWM Protocol. PWM Core: A 172-modality engine for verified scientific computing. <https://github.com/integritynoble/pwm>, 2026.
- Arnaud A. A. Setio, Alberto Traverso, Thomas de Bel, et al. Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge. *Medical Image Analysis*, 42:1–13, 2017.
- Authors to be confirmed. PWM-LDCT: A multi-vendor paired-dose CT benchmark dataset. Manuscript in preparation, target *Nature Scientific Data*, 2027a.
- Authors to be confirmed. Signal-equivalence: A probabilistic framework for comparing reconstruction methods across reduced-signal imaging modalities. Manuscript in preparation, target *Nature Methods*, 2027b.
- Authors to be confirmed at submission. Diffusion models in low-dose CT: A review. *Medical Image Analysis*, 2024. Placeholder citation; replace with the chosen diffusion baseline reference.
- Dayang Wang, Fenglei Fan, Zhan Wu, Rui Liu, Fei Wang, and Hengyong Yu. CTformer: Convolution-free token2token dilated vision transformer for low-dose CT denoising. *Physics in Medicine and Biology*, 68(6): 065012, 2023.
- Ge Wang, Jong Chul Ye, and Bruno De Man. Deep learning for tomographic image reconstruction. *Nature Machine Intelligence*, 2(12):737–748, 2020.

- Qingsong Yang, Pingkun Yan, Yanbo Zhang, et al. Low-dose CT image denoising using a generative adversarial network with Wasserstein distance and perceptual loss. *IEEE Transactions on Medical Imaging*, 37(6): 1348–1357, 2018.
- Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.